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MASTERS OF COMPUTER APPLICATIONS

MAIN PROJECT

22-382-0401

AI DevOps Commander: An Autonomous DevOps Agent with GitHub Actions-Based
Deployment Support

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ABSTRACT

Modern software delivery requires frequent deployments, yet DevOps workflows remain complex and error-prone due to manual steps, fragmented tooling, and inconsistent adherence to playbooks. This project presents **AI DevOps Commander**, a local-first DevOps automation system where users issue human-language commands (e.g., “Deploy v1.6 to staging and run tests”) through a web dashboard. A LangGraph-based interpreter converts instructions into a structured plan and enforces a **mandatory plan preview + approval gate** before any execution.

The system supports **local deployments** (Git/Docker/approved scripts only) and **GitHub Actions**–based deployments (workflow dispatch + run monitoring) using sandboxed adapters rather than unrestricted shell access. A watchdog agent evaluates logs, exit codes, and monitoring metrics (e.g., Prometheus) to detect failures and trigger policy-based rollback to a last-known-good version. A RAG layer validates plans against SOPs stored in a vector database (Chroma), improving operational correctness. The architecture combines a Next.js frontend with a FastAPI backend (Redis + RQ) and is designed to scale from SQLite to Postgres and from local execution to cloud/Kubernetes.